

BULLHORN COULEE RESERVOIR, AB, Canada

WMO: 718390

Lat: 49.295N

Lon: 113.359W

Elev: 1115

StdP: 88.63

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -29.0	(c) -25.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.8	(c) 30.1	(d) N/A	(e) 28.2	(f) N/A	(g) 26.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-34.9	33.1	3.9	2.2	-37.7	34.7	-40.0	36.0	-42.2	37.3	-45.0	38.9
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.1	-5.7	-0.4	4.4	9.9	14.1	17.6	17.0	12.3	5.7	-0.5	-6.8		
	DBStd	10.51	8.39	8.77	7.01	5.09	4.26	3.20	3.01	3.13	4.47	5.53	7.41	8.82		
	HDD10.0	2481	467	439	322	177	55	4	0	1	28	151	317	520		
	HDD18.3	4822	726	673	579	419	262	132	49	63	184	392	565	778		
	CDD10.0	754	0	0	1	7	52	127	236	217	96	17	2	0		
	CDD18.3	54	0	0	0	0	1	4	27	20	2	0	0	0		
	CDH23.3	1294	0	0	0	4	31	99	517	485	153	4	0	0		
	CDH26.7	352	0	0	0	0	2	18	157	144	31	1	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	548	27	24	34	49	76	107	47	46	47	33	31	28		
	PrecMax	844	43	49	64	94	166	323	173	118	135	79	61	59		
	PrecMin	343	8	6	12	17	9	24	3	3	5	6	10	9		
	PrecStd	120	9	11	13	17	38	67	38	28	29	19	13	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.4	15.0	23.0	26.3	29.6	33.0	32.5	30.0	22.5	17.5	10.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	8.1	9.4	12.5	18.7	23.9	25.8	30.4	30.0	27.2	19.3	13.4	7.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	6.5	7.1	10.7	15.7	21.6	24.0	28.3	28.2	25.0	16.6	10.9	5.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	5.2	5.2	8.6	12.9	18.9	22.0	26.3	26.1	22.1	14.0	8.7	4.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		10.0	11.0	10.9	11.5	12.1	12.3	14.8	15.6	14.1	11.7	10.3	10.3		
	5% DB	MCDBR	9.4	11.0	12.8	16.9	16.9	16.2	18.6	20.3	20.7	15.5	12.1	10.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.259	0.279	0.328	0.336	0.330	0.335	0.352	0.314	0.281	0.257	0.231		
	taud		2.409	2.407	2.445	2.338	2.394	2.456	2.448	2.386	2.466	2.542	2.495	2.423		
	Ebn at Noon		839	909	944	919	922	928	917	885	890	874	822	812		
	Edh at Noon		62	79	91	114	113	107	106	108	89	69	57	53		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.24	3.48	4.80	5.72	6.40	7.11	5.67	4.03	2.50	1.39	0.99		
	RadStd		0.10	0.17	0.30	0.32	0.50	0.49	0.39	0.35	0.33	0.24	0.11	0.09		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page